Mood-Based Music Recommendations with AI

An intelligent system that detects user emotions from text and delivers personalized Spotify playlists — automatically, instantly, and effortlessly.

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The Problem with Manual Music Discovery

Finding the right music for your current mood is a surprisingly friction-filled experience. Users must articulate what they want, browse manually, and hope they land on something that fits — all while the moment passes.

Time-Consuming

Manual searching disrupts the experience and breaks flow.

Emotionally Misaligned

No existing system reads emotional context automatically.

High Effort, Low Precision

Users often settle for "good enough" rather than the perfect fit.

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The Solution: AI-Driven Mood Detection

Users simply describe how they feel in plain text. The system takes it from there — detecting the emotion, querying Spotify, and returning a curated playlist in seconds.

Text Input

User describes their mood in natural language.

AI Emotion Detection

Llama3 via Ollama classifies the emotional state.

Playlist Fetch

Spotify API returns mood-matched playlists instantly.

Instant Result

Mood label, playlist name, and link delivered to the user.

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System Architecture

Enter Mood

Send Request

n8n Webhook

LLM Detects

Fetch Playlists

The pipeline is fully event-driven: n8n orchestrates the workflow, Ollama handles local LLM inference for mood classification, and the Spotify API closes the loop with real playlist data — all without manual intervention.

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Tech Stack at a Glance

Frontend

HTML, CSS, JavaScript — lightweight and accessible interface for mood input and result display.

Orchestration

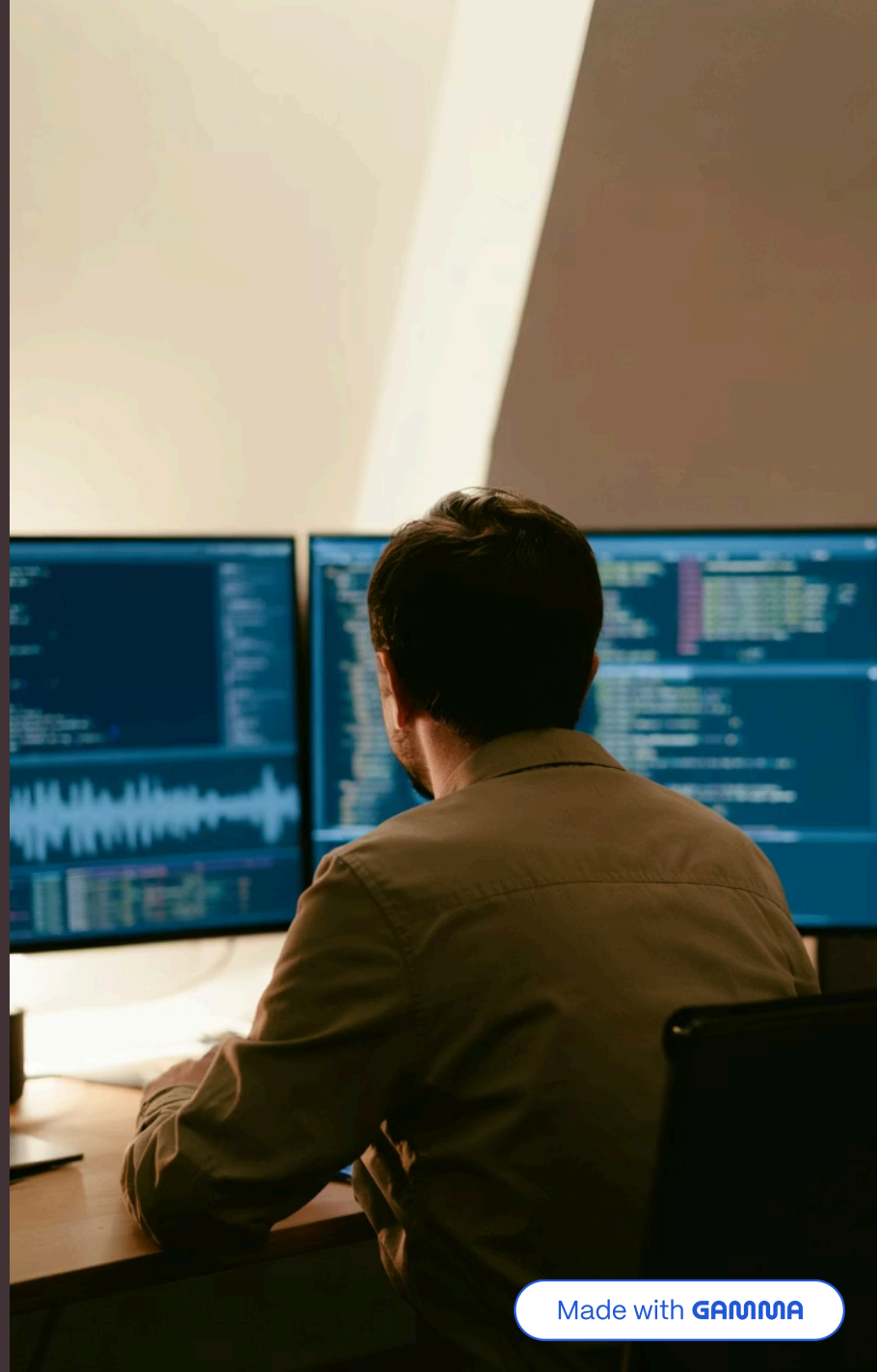
n8n Webhook — no-code workflow automation that routes requests and manages the pipeline logic.

AI Model

Ollama + Llama3 — locally-run LLM that classifies free-form mood text into structured emotion labels.

Music API

Spotify API — fetches curated playlists based on the detected mood category in real time.



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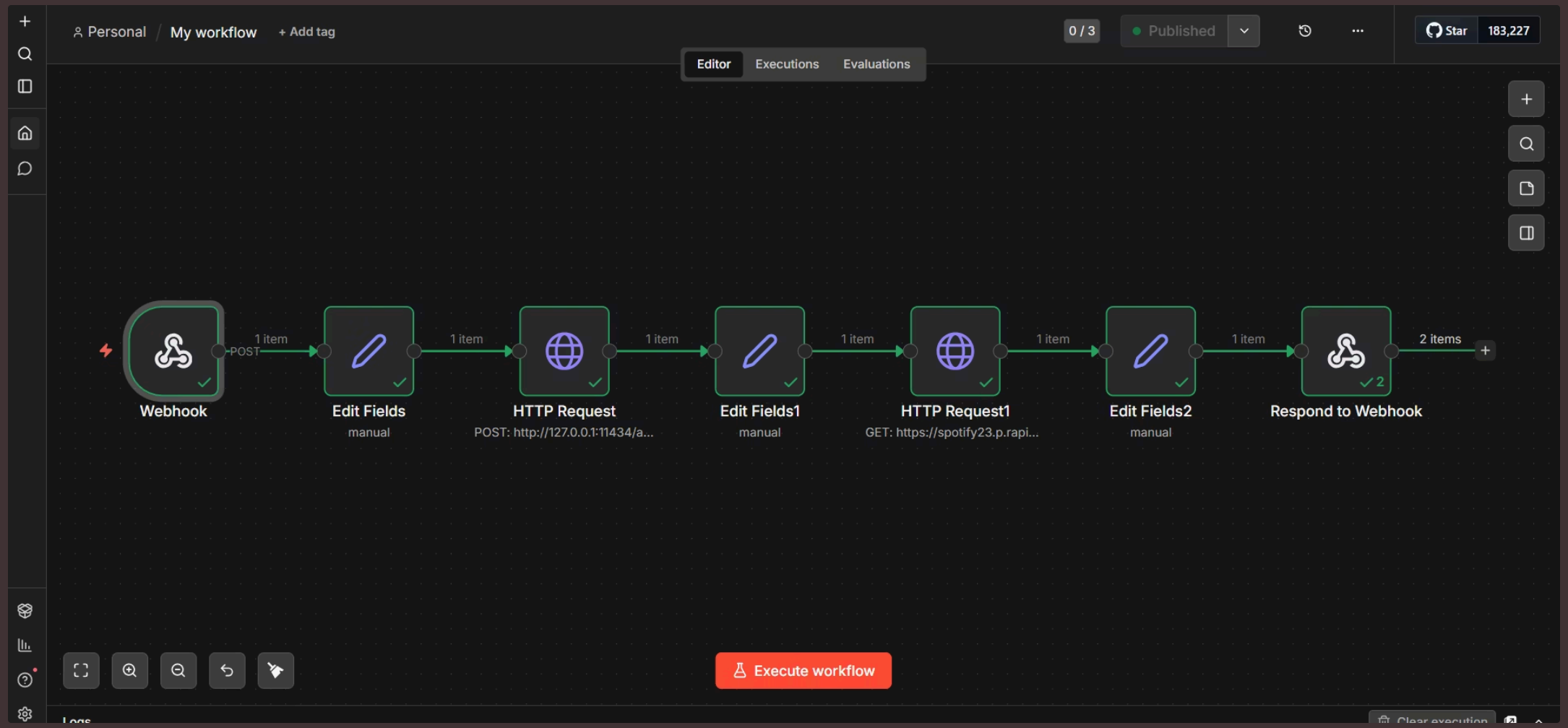

n8n WORKFLOW

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Overview | Overview | POST Get da | GET New Requ | GET New Requ | POST http://l | POST http://l | POST http://l | No environment

COLLECTIONS
- My Collection
- New Collection
  - GET New Request
  - GET New Request

ENVIRONMENTS

No items in this panel

+ Create

My Collection > **Get data**

Save | Share

POST | http://localhost:5678/webhook/mood | Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Tests | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2   "mood_text": "I feel Happy today"
3 }
```

Body | Cookies | Headers 8 | Test Results 1/1 | 200 OK | 17.48 s | 612 B | Save Response

{} JSON | Preview | Visualize

```
1 {
2   "mood": "Happy",
3   "playlist_name": "Telugu happy mood songs",
4   "playlist_url": "https://open.spotify.com/playlist/4jW37umAGFKr2oQRAk5pAe"
5 }
```

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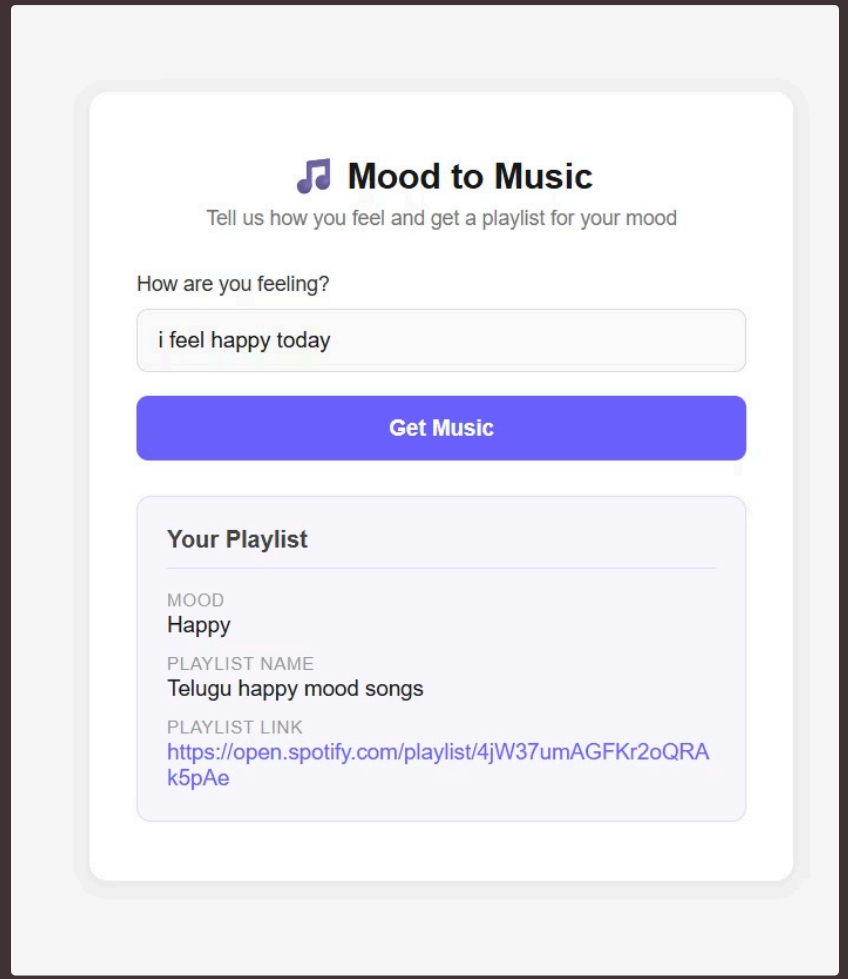

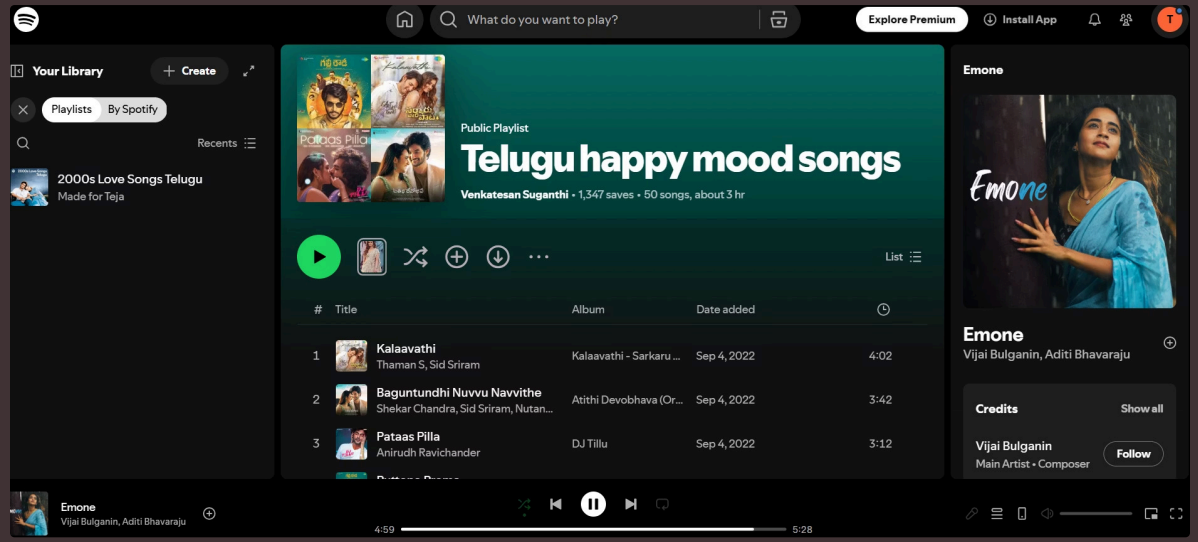

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Future Scope

The current system is a strong foundation. Planned enhancements will make it smarter, more personal, and platform-agnostic.

Advanced AI Models

Fine-tuned emotion classifiers for higher accuracy and nuance.

Voice Input

Speech-to-text integration for hands-free mood entry.

Personalization

Recommendation history and user preference profiles.

Multi-Platform

Expand beyond Spotify to YouTube Music and Apple Music.

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Conclusion

This project is a working proof-of-concept for **emotionally-aware AI applications** — combining local LLM inference, workflow automation, and third-party API integration into a seamless user experience.

AI + Automation

Demonstrates practical LLM use in real product workflows.

Zero Manual Effort

From mood text to playlist link with no user intervention.

Extensible by Design

Architecture scales to new models, inputs, and platforms.

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